Cross-dataset transfer of a Boltz raw-Δz ΔΔG predictor

Experiment 05 — Tsuboyama → FireProt. ddG_with_Boltz project · Generated 2026-07-18 · raw-Δz features, MLP (5-seed ensemble) primary model.

> A ΔΔG predictor trained on the entire **Tsuboyama** mega-scale folding dataset (12,359 mutations) transfers — with **no refitting** — to the independent **FireProt** dataset (1,543 mutations / 85 proteins, **zero protein overlap**): pooled Pearson **r = 0.621**, Spearman **ρ = 0.684**, per-protein **median r = 0.67**. The Boltz raw-Δz signal is not an artifact of one dataset. Its ceiling is **magnitude, not ranking**: predictions span only ~40% of the true ΔΔG spread (fit slope 0.26), so the model ranks and triages well but under-predicts absolute effect sizes on out-of-range mutations.

1. Objective

Every generalization test so far (experiments 01, 06) splits *within* the Tsuboyama corpus. Such holdouts can still ride on quirks shared by a single data source — Tsuboyama is one high-throughput folding assay on mostly small/designed domains. The decisive question for a usable predictor is whether the signal carries to an **independently curated** dataset: different proteins, a different assay, and a different label provenance. We test transfer to **FireProt** (literature-derived stability measurements, ≤200 aa here).

2. Data & methods

- **Train:** all 12,359 Tsuboyama mutations (`tsuboyama_bench_fast/rawz_features.parquet`), **256 raw-Δz features** (mutant−WT difference of the Boltz-2 pair track *z*: `zdiag_*` 128 + `zpool_*` 128).
- **Test:** all 1,543 FireProt mutations / 85 proteins (`fireprot_le200/features_summary.parquet`), the *same* 256 features.
- **Model:** the benchmark pipeline `SimpleImputer(median) → StandardScaler → estimator`, fit once on Tsuboyama and applied unchanged to FireProt. Primary estimator: a **5-seed MLP ensemble** (project default since exp 06); HistGradientBoosting (HGB) reported alongside.
- **Entry point:** `python -m ddg.evaluation.transfer --train … --test … --model mlp`.
- **Independence:** the train and test protein sets have **zero `wt_id` overlap**. FireProt's ΔΔG range [-13.7, 12.0] is much wider than Tsuboyama's [-2.7, 5.7].
- **Sign convention:** FireProt (`ddG`) and Tsuboyama (`ddg`) share the same sign (positive = destabilizing; both ~75% positive), so predictions are used as-is (no flip).

3. Results

Metric	MLP (primary)	HGB
Pooled Pearson r	**0.621**	0.607
Pooled Spearman ρ	**0.684**	0.675
Pooled RMSE	1.41	1.45
Pooled MAE	0.86	0.89
Per-protein r — mean	0.488	0.531
Per-protein r — median	**0.668**	—
Proteins scored	76 / 85	76 / 85

Table 1. Pooled and per-protein transfer metrics (n = 1,543 mutations / 85 proteins). ΔΔG in kcal/mol.

MLP and HGB are within noise of each other — the same "representation, not model" result as experiment 06. 70% of proteins score r > 0.5 and 47% score r > 0.7; the per-protein *mean* (0.49) sits well below the *median* (0.67) because a handful of proteins transfer poorly (Fig. 2).

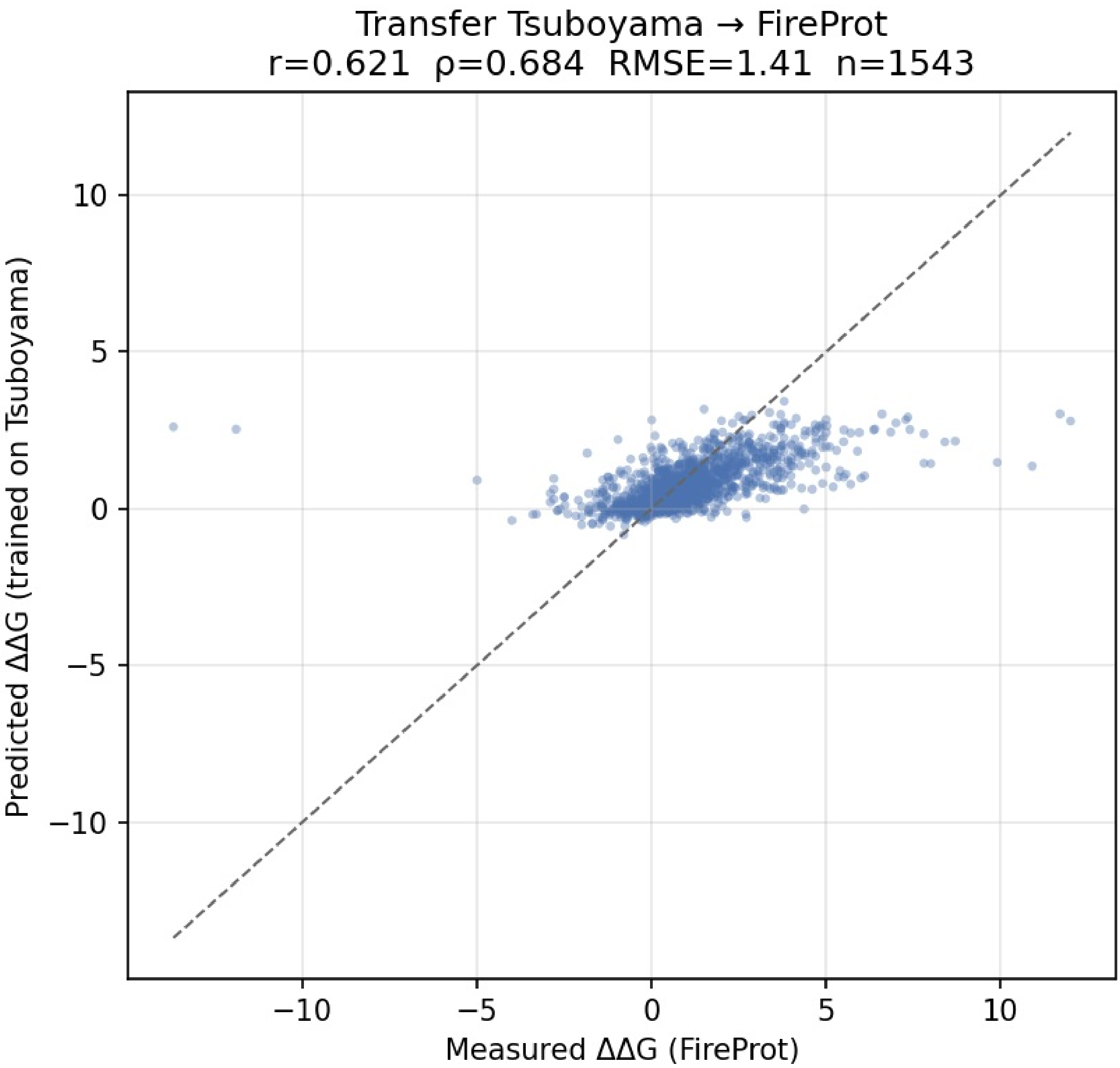

Figure 1. Predicted (trained on Tsuboyama) vs measured (FireProt) ΔΔG for all 1,543 mutations. Dashed line is y = x. The cloud is strongly correlated but far flatter than the diagonal: predicted ΔΔG concentrates in ~[0, 3] while measured values span [-13.7, 12.0] — the magnitude compression quantified in §4.

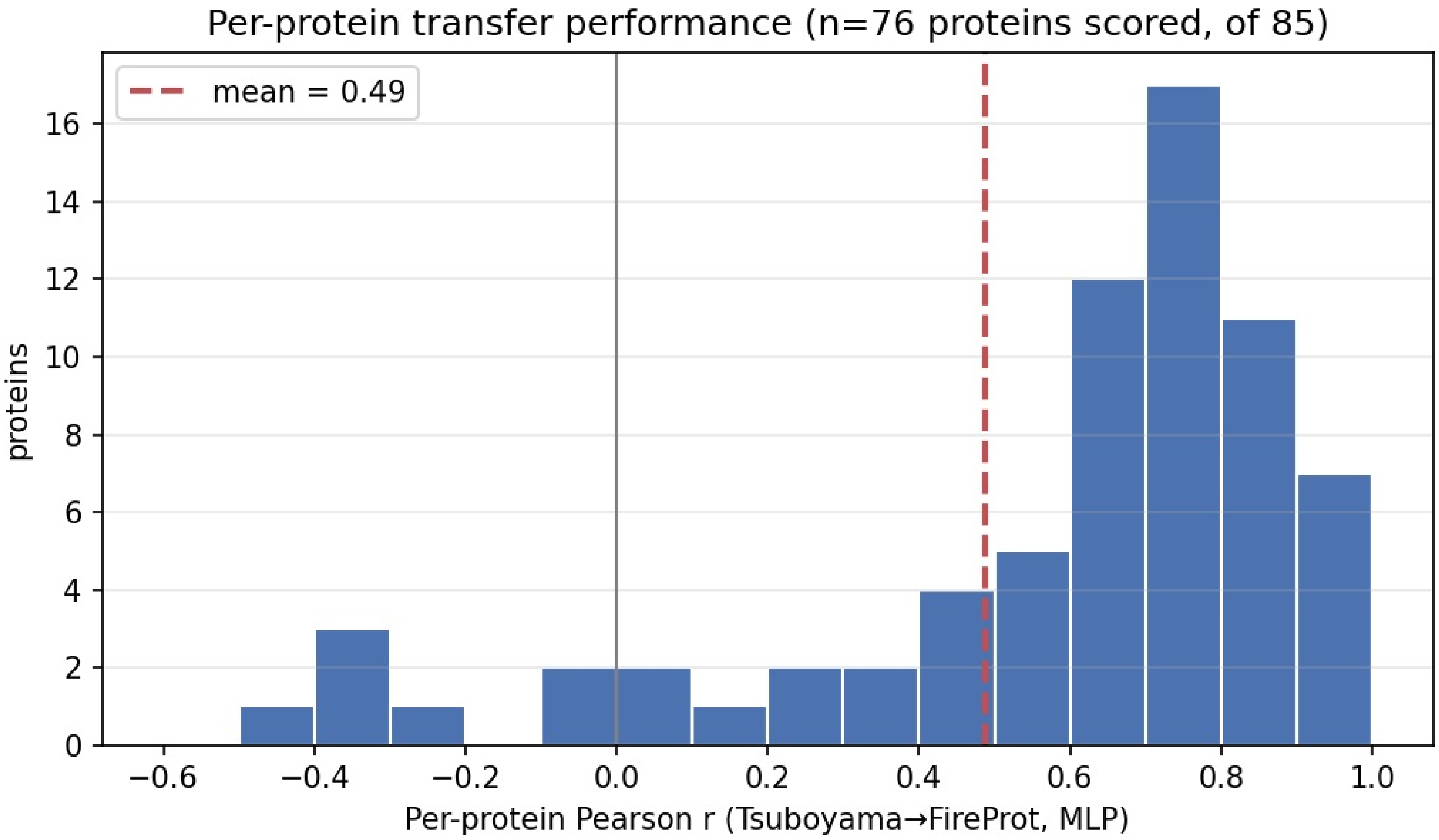

Figure 2. Distribution of per-protein Pearson r (76 of 85 proteins scored; the rest have <2 mutations or constant ΔΔG). Concentrated at 0.6–1.0 (median 0.67); the mean 0.49 is pulled down by a few poorly-transferring proteins.

Protein	n	Pearson r	Spearman ρ	RMSE
P08463	14	0.895	0.664	0.853
P01053	49	0.877	0.900	0.896
P02417	26	0.864	0.875	0.996
P0AA04	12	0.847	0.792	0.930
P00648	114	0.835	0.740	1.663
P61823	46	0.821	0.855	2.551
… 65 proteins between …				
3PG0	10	-0.447	-0.148	1.061
P11540	9	-0.359	-0.452	1.006
P37001	19	-0.284	-0.319	0.658
P05230	15	0.071	0.386	0.873
P00282	18	0.094	0.434	3.126

Table 2. Best- and worst-transferring proteins (≥8 mutations each).

4. The ceiling: magnitude, not ranking

The predictor *ranks* mutations well (ρ = 0.68) but severely **under-predicts magnitude**. The predicted-vs-measured fit slope is **0.26**, and predicted ΔΔG has SD 0.72 against a measured SD of 1.72 — only ~42% of the true spread. FireProt's wide range makes this starker than any within-Tsuboyama split: the model regresses toward the mean on the destabilizing/stabilizing tails it never saw in training. This is the same regression-to-the-mean weakness isolated in experiment 02 (extrapolation) and confirmed model-independent in 06 — a property of the features/objective, not of the estimator. **Practical consequence:** use for ranking, prioritization, and triage — not for absolute ΔΔG on mutations outside the training range.

5. Corpus completeness (provenance note)

Reaching the full corpus required two fixes, both logged in `status.md`:

- The originally committed feature table held only **773 / 54 proteins** — the `slim` step had run on a half-finished predict array, and `delete_raw: true` had deleted the raw embeddings, preventing a cheap resume. A clean full predict re-run reached **1,514 / 82**.
- The remaining 29 mutations belonged to **3 proteins** (3PG0/ThreeFoil, 2IMM, 1YYX) that FireProt lists with only a `pdb_id`, no `uniprot_id`; the adapter keyed `wt_id` on `uniprot_id` alone and dropped them. A `pdb_id` fallback recovered all 29 → **1,543 / 85** (the corpus used here).

Protein	n	Pearson r	Spearman ρ	RMSE
1YYX	14	0.687	0.644	1.989
3PG0	10	-0.447	-0.148	1.061
2IMM	5	0.789	0.821	0.487

Table 3. Transfer for the 3 recovered (UniProt-less) proteins.

6. Provenance

Train config `experiment_configs/tsuboyama_bench_fast.yaml` · train features `data/processed/tsuboyama_bench_fast/rawz_features.parquet` · test config `experiment_configs/fireprot_le200.yaml` · test raw `data/raw/fireprot_le200.csv` · test features `data/processed/fireprot_le200/features_summary.parquet` · transfer code `ddg/evaluation/transfer.py` (commit 5e55812) · outputs `data/processed/fireprot_le200/transfer_from_tsuboyama{,_hgb}/`. Cluster feature extraction: SLURM jobs 212209 (prepare) → 212210 (predict, 8 shards, 1,628 structures) → 212211 (slim) → 212212 (features), all COMPLETED. `data/processed/` is gitignored; only configs, raw data, and `results/` are committed.